Scouting and Talent Evaluation

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HES 3000-002, Sport Performance Analytics

Analytics Department Project

- You are on an analytics team during draft season. There are 60 prospects, 20 metrics each. A week to decide. What do you do?
    - Put a plan together: Write down your first 3 steps
    - Get in groups of 3, share with each other

Getting specific

- “Sort by the best stat” → *which* stat?
- “Best” → compared to *whom*?
- “Good enough” → good enough for *what*?
- “Look at everything” → how do I *combine* it?

Outline:

- KPI → Group → Method → Composite

1. Which KPIs matter?

How to choose a KPI

- Understand what each measure







			Regular Season						Playoffs				
Season	Team	Lge	GP	G	A	Pts	PIM	+/-	GP	G	A	Pts	PIM
2023-24	Chicago Steel	USHL	49	5	4	9	38	-8	3	0	0	0	2
2024-25	Chicago Steel	USHL	2	0	0	0	0	-1					
2024-25	Sioux Falls Stampede	USHL	50	16	17	33	30	1	4	1	1	2	0
2025-26	London Knights	OHL	37	12	22	34	23	11	--	--	--	--	--
2025-26	Barrie Colts	OHL	29	15	17	32	24	19	20	11	11	22	6
2026-27	Ohio State University	Big-10	4	1	2	3	0	2					

*KPI = Key Performance Indicator

How to choose a KPI

- Understand what each measure

category	metric	defined	mean	max	min
aerobic fitness	vo2max_ml_kg_min				
agility	pro_agility_left_sec				
agility	pro_agility_right_sec				
anaerobic fitness	anaerobic_peak_power_watts_kg				
anaerobic fitness	anaerobic_fatigue_index_pct				
anthropometric	wingspan_in				
lower body power	horizontal_jump_in				
lower body power	isokinetic_squat_lb_per_lb_bw				
lower body power	force_plate_no_arm_jump_in				
lower body power	force_plate_vertical_jump_in				
speed	sprint_10m_sec				
upper body power	bench_press_power_watts_kg				
upper body power	right_hand_grip_lb				
upper body power	left_hand_grip_lb				
upper body power	pull_ups_reps				

How to choose a KPI

- Understand what each measures
- Identify KPIs that are important to your team
  - Coach’s directive
  - Status quo
  - Analytical methods

*KPI = Key Performance Indicator

Real Analytics is Collaborative

Analysts rarely make decisions start to finish.

- Coaching staff may:
  - Describe outcome
    - “I want a winger who plays with pace, doesn't cough up the puck at the blue line, and can kill penalties.”
    - Use analytics to find metrics who fits this
  - Prescribe KPI
    - “I want to see players with good xGF%, speed and agility.”
    - Use analytics to find prospects who excel in this.

KPI Status Quo – use or drop?

- Many metrics are already known to be important:
  - tradition, past research, what other teams share
- These are a sensible starting point
  - if every team uses the same KPIs, it is not a competitive advantage
- Edge can come from:
  - Checking whether a known KPI holds up in ***your*** **data**
  - Matching metrics to **your team's needs** (needs aren't the same across teams)
  - **Finding what the standard KPI list misses**

Analytics to find KPIs

- Not every number is useful
  - Some are **signals:** tell us something real about future success
  - Some are **noise:** random, redundant, or irrelevant metrics

- KPI filters: Relevant? Distinct?

Start with the outcome

- You can’t pick predictors until you define “success”
  - Eg: minutes played in years 1-3, win shares, still in the league after 4 years, role (starter or bench)

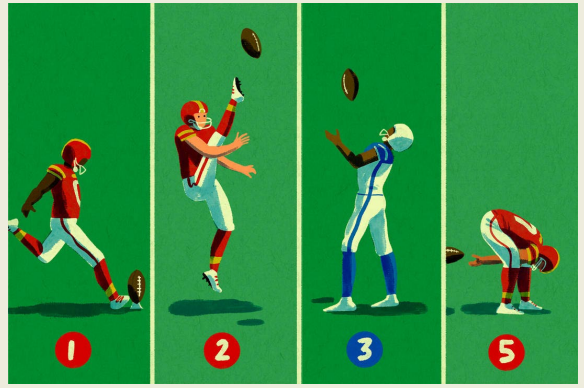
- Different outcomes → different “best” KPI

- How would you define success for …
  - NFL Player
  - NHL Player
  - WNBA Player



Define success

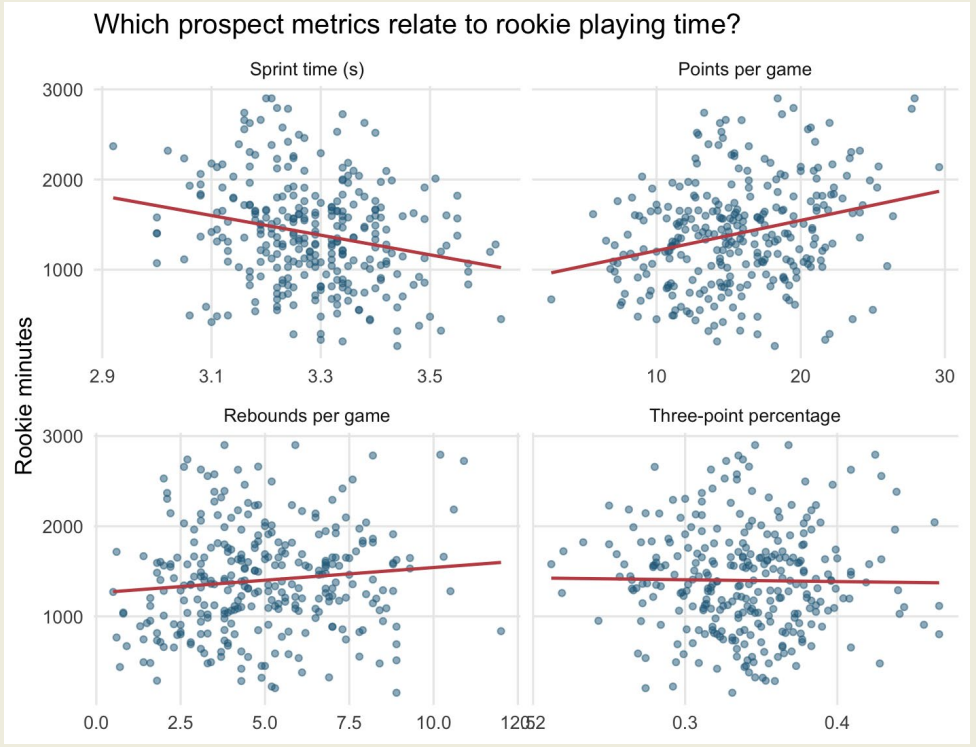
- Would these qualities change if the question was
  - NFL QB vs WR vs DB?
  - NHL Goalie vs Center vs Wing?
  - WBNA PG vs F vs C?



- Would these qualities change if the question was
  - NFL QB for the Bears vs the Broncos?

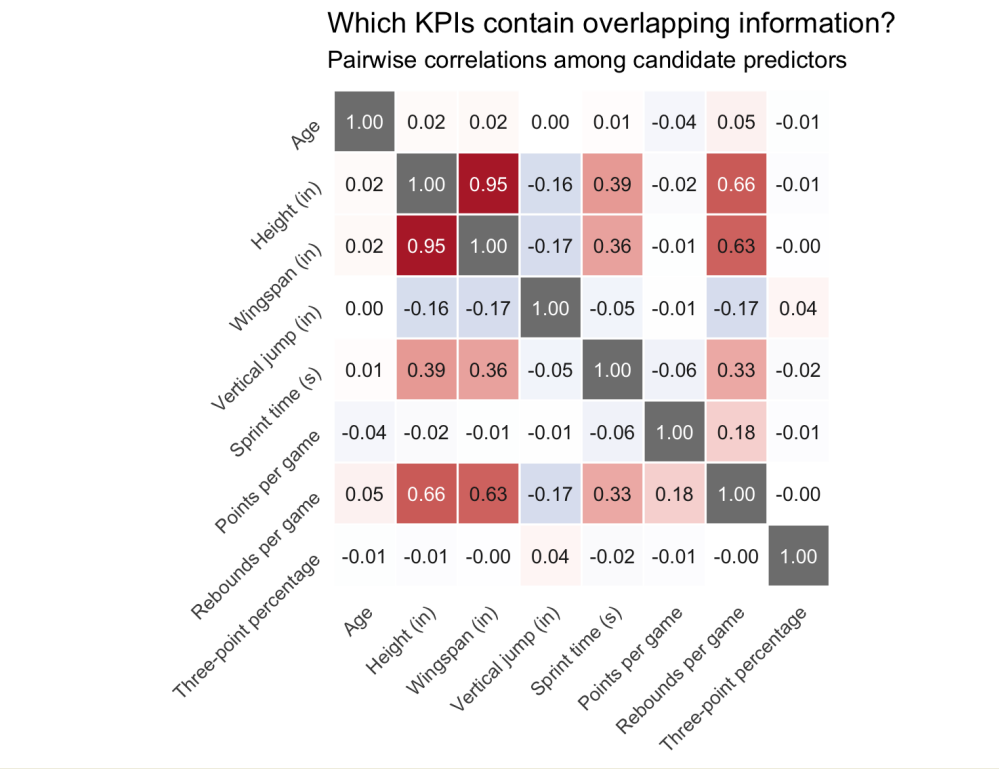
Filter 1: Is it relevant?

- Correlate KPIs from historical datasets with success indicators
- A weak correlation can mean the metric is unrelated *or* just noisy.



Filter 2: Is it distinct?

- Correlate KPIs with each other
  - If two metrics at r ≈ 0.9 are telling us roughly the same thing
  - Redundant metrics double-count one quality



Common traps

- **Chance:**
  - The more metrics you screen, the greater the chance that some appear relevant by luck.
- **Survivors:**
  - Looking only at drafted players can distort relationships and limits what we can conclude about all prospects.
- **Correlation ≠ causation, and circularity:**
  - Outcome can reflect team decisions, not only individual performance
- **Look before you correlate**:
  - linear correlation analysis will only test linear relationships

2. Compared to whom?

Compared with whom?

- A number means nothing alone:
  - Is a 58 mL/kg/min VO2max good?
- Compared to…
  - the whole draft class?
  - Position group?
  - Conference?
  - Current pros?
- The group you pick changes the answer



How does your VO2 max measure up?

Treadmill-based VO2 max classifications by age and sex

Men	% Rank	20-29	30-39	40-49	50-59	60-69	70-79	80-89
Superior	95th	63.5	58.8	54.6	47.6	40.6	34.0	24.1
Excellent	80th	54.5	50.0	45.2	38.3	32.0	25.9	21.4
Good	60th	49.0	43.4	37.9	31.8	26.5	22.2	18.4
Fair	40th	43.6	37.0	32.4	26.9	22.8	19.1	16.6
Poor	20th	35.2	29.8	26.7	22.2	18.5	15.9	14.8

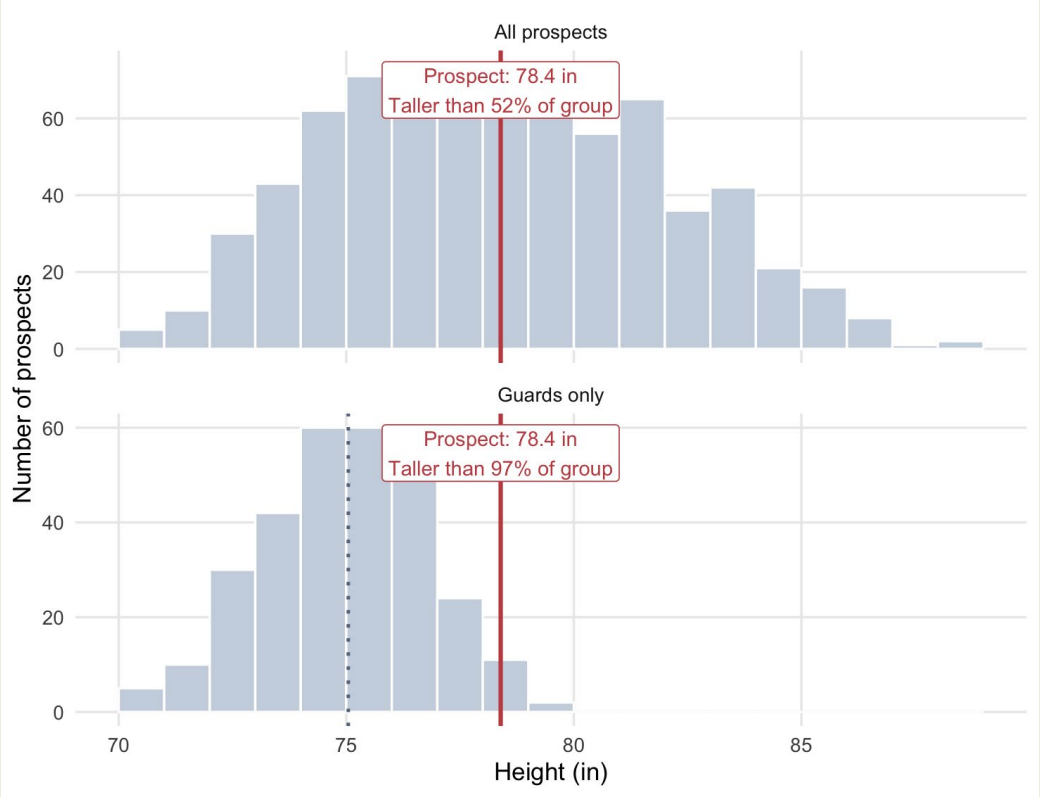
Women	% Rank	20-29	30-39	40-49	50-59	60-69	70-79	80-89
Superior	95th	52.1	45.9	41.1	36.0	30.0	24.3	22.7
Excellent	80th	44.8	37.0	33.0	28.4	24.3	20.8	18.4
Good	60th	39.0	31.0	27.7	24.6	20.9	18.3	16.0
Fair	40th	34.0	26.4	23.9	21.5	18.3	16.2	14.7
Poor	20th	27.2	21.9	19.7	18.5	15.4	14.0	12.6

Source: American College of Sports Medicine's Guidelines for Exercise Testing and Prescription, 12th edition

Cleveland Clinic

The comparison group changes the interpretation.

- Selecting the appropriate comparison group is important
- One athlete can look average in one group, and elite in a sub-group

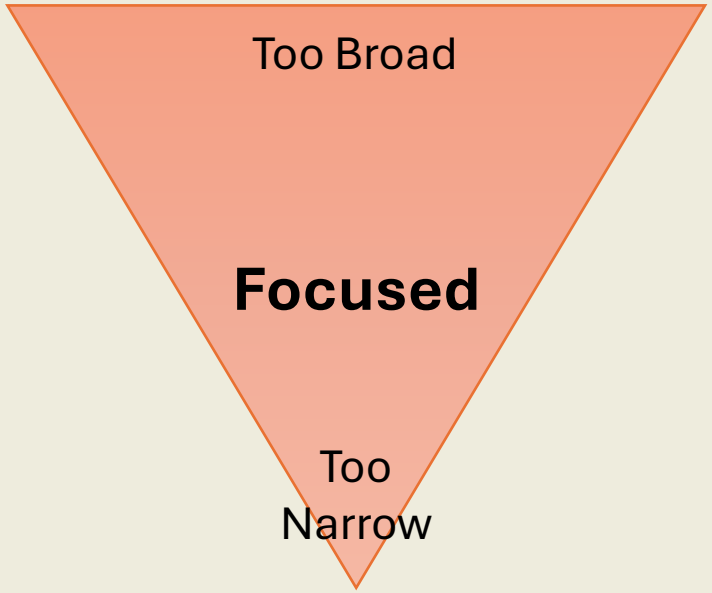


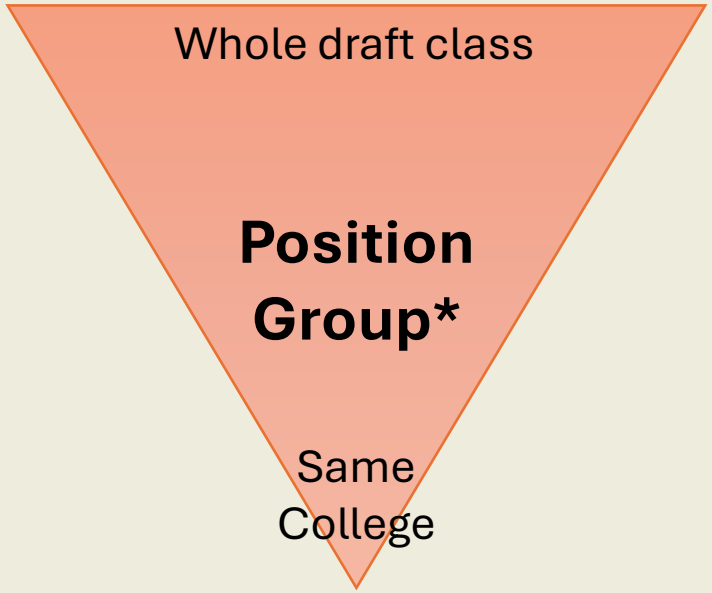
Choosing a comparison group

- **General population**: …
- **Whole draft class:** simple, but mixes roles
- **Same position:** fairer for position-specific traits
- **Same conference / level of competition:** provide a more similar competition context
- **Same age or class year:** accounts for development

Choosing a comparison group

- Trade-off: more specific = more relevant, but smaller sample





* Most appropriate comparison depends on context

Summary

- Define success
- Choose relevant, distinct KPIs
- Choose the comparison group
- Relative comparisons vs benchmark comparison

- Return to answers for opening activity
  - Which step would you add or change?

- Up Next: How do we *express* the comparison, and how do we *combine* metrics?